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Bronchoalveolar Lavage Lymphocytes in the Diagnosis of Hypersensitivity Pneumonitis among Patients with Interstitial Lung Disease

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Abstract

Rationale: Hypersensitivity pneumonitis (HP) is an interstitial lung disease (ILD) characterized by inflammation and/or fibrosis in response to an inhalational exposure.

Objectives: To determine the value of bronchoalveolar lavage (BAL) fluid lymphocyte cellular analysis in the detection of HP among patients with newly detected ILD.

Methods: This systematic review was undertaken in the context of development of an American Thoracic Society, Japanese Respiratory Society, and Asociación Latinoamericana del Tórax clinical practice guideline. The clinical question was, "should patients with newly detected ILD undergo BAL fluid lymphocyte analysis to diagnose HP?" MEDLINE, EMBASE, and the gray literature were searched through October 2019. Studies that reported the percentage of BAL fluid lymphocytes for various ILDs were selected for inclusion. Meta-analyses compared the mean percentage of BAL fluid lymphocytes among patients with HP with that among patients with idiopathic pulmonary fibrosis (IPF) or sarcoidosis. The sensitivity and specificity by which various percentages of BAL fluid lymphocytes distinguish HP from IPF and sarcoidosis were also evaluated.

Results: Eighty-four articles were selected. No randomized trials or observational studies were identified that compared BAL fluid lymphocyte analysis with no BAL fluid lymphocyte analysis in patients with ILD. Included studies were case series describing BAL fluid cell differentials in patients with various ILDs. The percentage of BAL fluid lymphocytes was significantly higher in both fibrotic and nonfibrotic HP compared with IPF. Similarly, the percentage of BAL fluid lymphocytes was significantly higher in both fibrotic and nonfibrotic HP compared with sarcoidosis. A threshold of 20% BAL fluid lymphocytes distinguished fibrotic HP from IPF with a sensitivity and specificity of 69% and 61%, respectively, and nonfibrotic HP from IPF with a sensitivity and specificity of 95% and 61%, respectively. It distinguished fibrotic HP from sarcoidosis with a sensitivity and specificity of 69% and 26%, respectively, and nonfibrotic HP from sarcoidosis with a sensitivity and specificity of 95% and 26%, respectively.

Conclusions: The percentage of BAL fluid lymphocytes is higher in HP than IPF or sarcoidosis. However, a threshold that distinguishes HP from IPF or sarcoidosis with both high sensitivity and high specificity was not identified.

Keywords: bronchoalveolar lavage; lymphocytes; diagnosis; hypersensitivity pneumonitis; interstitial lung disease

(Received in original form May 4, 2020; accepted in final form June 30, 2020)

A systematic review that informed the American Thoracic Society/Japanese Respiratory Society/Asociación Latinoamericana del Tórax Guideline on the diagnosis of hypersensitivity pneumonitis.

Supported by the American Thoracic Society.

Author Contributions: S.P. and M.T.K.: design of the project, preparing and validation of search strategy, searching bibliographic databases, title abstract screening, full text screening, data extraction, data analysis, and drafting of manuscript. H.A.C., A.C., J.D.-M., A.D., and A.R.J.: data analysis and drafting of manuscript. S.L.K.: preparing and validation of search strategy, and searching bibliographic databases. G.R.: conception and design of the project, verification and supervision over all parts of the project, and drafting of manuscript. K.C.W.: conception and design of the project, preparing and validation of search strategy, searching bibliographic databases, title abstract screening, full text screening, data extraction, data analysis, drafting of manuscript, and supervision over all parts of the project.

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This article has a related editorial.

This article has an online supplement, which is accessible from this issue's table of contents at www.atsjournals.org.

Ann Am Thorac Soc Vol 17, No 11, pp 1455–1467, Nov 2020
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DOI: 10.1513/AnnalsATS.202005-420OC
Internet address: www.atsjournals.org

Hypersensitivity pneumonitis (HP) is an interstitial lung disease (ILD) characterized by immune-mediated inflammation with or without fibrosis in response to inhalational exposure to various inciting agents (1). The incidence of HP varies with geographic location. In the United States, the 1-year cumulative incidence of HP ranges from 1.28 to 1.94 cases per 100,000 persons (2). More than half of these cases are related to chronic HP (2).

Various definitions, classifications, and diagnostic criteria have been proposed for HP (3–7), but they have failed to gain widespread acceptance. A new definition and updated classification and diagnostic criteria for HP are needed to standardize clinical care and facilitate research (8). Bronchoalveolar lavage (BAL) lymphocytosis has been proposed as a diagnostic criterion for HP (3, 4, 9). However, the ability of BAL fluid lymphocyte analysis to distinguish the nonfibrotic and fibrotic types of HP from various ILDs is unknown, as is the optimal threshold BAL lymphocyte count to diagnose HP (10–12).

This systematic review was performed in the context of development of an American Thoracic Society, Japanese Respiratory Society, and Asociación Latinoamericana del Tórax guideline on the diagnosis of HP (13). The goals of the systematic review were to answer the question, "should patients with newly detected ILD in a chest radiograph or computed tomographic (CT) scan of the chest, with or without a history of exposure capable of causing HP, undergo BAL fluid lymphocyte analysis to diagnosis HP?" and to identify a threshold BAL lymphocyte count that distinguishes HP from other ILDs.

Methods

This review was performed in accordance with the standards described in the Cochrane Handbook of Systematic Reviews of Diagnostic Test Accuracy (14). The systematic review was not registered because it was performed in the context of guideline development rather than *de novo* and was initiated before the requirement that such systematic reviews be registered.

Research Questions

The population, intervention, comparator, outcome format was used to develop the question, "should patients with newly detected ILD in a chest radiograph or CT scan of the chest, with or without a history of exposure capable of causing HP, undergo BAL fluid lymphocyte cellular analysis to diagnosis HP?" The population was adult patients who have newly detected ILD, the intervention was BAL fluid lymphocyte cellular analysis, the comparator was no BAL fluid lymphocyte cellular analysis, and the outcomes included diagnostic test characteristics. A secondary question was also posed: "is there a threshold percentage of BAL fluid lymphocytes that is diagnostic of HP?"

Literature Search

Potentially relevant systematic reviews were initially sought in the Cochrane Library and MEDLINE, but none were identified. A sensitive search strategy was therefore developed and used to search for potentially relevant studies in MEDLINE, EMBASE, and the gray literature (Table E1 in the online supplement). An initial search encompassed January 1946 through April 2019, which was updated to include through October 2019.

Study Selection

Two reviewers independently reviewed all studies for inclusion and exclusion criteria. Disagreements were resolved by discussion. The study selection process was established *a priori* (Table E2). Randomized controlled trials comparing BAL fluid lymphocyte cellular analysis with no BAL fluid lymphocyte cellular analysis in patients with newly detected ILD were initially sought. If none were found, observational studies comparing BAL fluid lymphocyte cellular analysis with no BAL fluid lymphocyte cellular analysis in patients with ILD were sought. If none were found, case series that enrolled patients with ILD who underwent BAL fluid lymphocyte cellular analysis were sought, including studies that reported the BAL fluid lymphocyte count or proportion in a given type of ILD and studies that compared the BAL fluid lymphocyte count or proportion in different types of ILD. Case series with fewer than 20 patients and non–English-language studies were excluded.

Data Extraction

A standardized form was developed for data extraction from the selected studies. Data were extracted independently by two investigators. Extracted information included the author, date, study location, study design, diagnostic criteria for various ILDs, number of participants, chest radiograph and CT scan findings, details of intervention, details of the comparator, and outcomes. Extracted data were reviewed and confirmed by a third reviewer.

Data Synthesis

To determine whether BAL can be used to distinguish HP from other ILDs, meta-analyses were performed using studies in which the mean percentage of BAL fluid lymphocytes in patients with HP was compared with that in patients with other ILDs. The meta-analyses used a random effects model in the Cochrane Collaboration Review Manager, version 5.3 (The Nordic Cochrane Centre). The mean difference (MD) and accompanying 95% confidence interval (95% CI) was the summary estimate. Subgroup analyses were performed for patients with fibrotic (chronic) HP, nonfibrotic (acute or subacute) HP, and nonspecified HP.

Statistical heterogeneity was measured using the I^2 test; an $I^2 \geqslant 75\%$, 50–75%, and 25–50% was considered severe, moderate, and mild, respectively. When heterogeneity was encountered, sensitivity analyses were performed to identify the contributing studies and then reasons for the heterogeneity were sought. The sensitivity analyses consisted of removing from the meta-analysis studies whose results appeared to be different from most studies and then reassessing the I^2 statistic. If the I^2 statistic improved, the full text of the removed studies was reviewed to determine if those studies were similar to each other and different from the others in a way that might explain the different results, such as enrolling a different population, performing the BAL differently, or measuring the lymphocyte count differently. If no cause was found, outliers were eliminated. Outliers were defined as studies whose confidence interval did not overlap the confidence interval of the pooled estimate. The estimates after elimination of the outliers are reported here, but results both before and after the elimination of outliers are shown in the figures.

To identify a threshold that distinguishes HP from other ILDs, we had hoped to find multiple studies conducted at various thresholds that could be pooled, or to perform an individual patient data meta-analysis to evaluate various thresholds.

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However, the former did not exist, and the latter proved infeasible. Therefore, we assumed the percentage of BAL fluid lymphocytes is normally distributed and created two normal distributions, one representing a disease-positive population (i.e., patients with HP) and the other representing a disease-negative population (i.e., patients with other ILDs). As the sample size increases, confidence in the normal distribution increases; therefore, to obtain a larger sample size, similar subgroups were combined. For example, to obtain a larger sample of patients with nonfibrotic HP, we combined all patients with nonfibrotic HP irrespective of their comparator. Studies previously identified as outliers were not used to calculate the mean of the normal distribution. IBM SPSS Statistic 24 software was used to obtain the normal distributions based on mean and standard deviation (SD). For BAL lymphocyte counts ranging from 0% to 100%, true-positive, false-negative, true-negative, and false-positive rates were determined at 5% increments, and a summary receiver operator curve was constructed using Microsoft Excel. The area under the curve (AUC) was calculated.

Quality of Evidence

Quality of evidence indicates confidence in the accuracy of the estimated effects. It was assessed using the Grading of Recommendations, Assessment, Development, and Evaluation approach. The Grading of Recommendations, Assessment, Development, and Evaluation approach makes a baseline assessment based upon study design: randomized trials are high-quality evidence, controlled observational studies are low-quality evidence, and case series are very low-quality evidence. After making a baseline assumption, five domains are assessed: risk of bias (internal validity), indirectness (external validity), inconsistency of estimates across the studies (heterogeneity), imprecision (wide CIs), and publication bias. If any of the domains were present, the quality of evidence is downgraded. The quality of evidence can also be upgraded if there is a large magnitude of effect, dose–response gradient, or if likely confounders would bias the study toward the opposite result.

To assess the risk of bias domain, the Newcastle-Ottawa scale and the Quadas-2 tool were modified to be applicable for case series. Potential risks of bias included not consecutively enrolling patients (i.e., selection bias), patients leaving the study (i.e., attrition bias), and not analyzing or reporting data collected (i.e., reporting bias). To assess for potential publication bias, funnel plots were created using RevMan (Figure E1).

Evidence Profile

Evidence profiles were developed that summarize the estimated effects and judgments made to rate the quality of evidence. During the development of the clinical practice guideline, the evidence profiles were presented to the guideline panel to inform their recommendations.

Results

The literature search identified 1,500 articles, of which 340 articles underwent full-text review (Figure 1). There were no randomized trials or observational studies comparing BAL fluid lymphocyte cellular analysis with no BAL fluid lymphocyte cellular analysis in patients with ILD. Therefore, 84 case series were selected that reported BAL fluid lymphocyte counts for HP and other ILDs (15–96) (Table E3). The series ranged in size from 19 to 618 patients. All of the series were retrospective, enrolled patients with HP, and reported BAL lymphocyte count retrospectively. The series used different criteria to diagnose HP. All used clinical and radiographic criteria, but some added serological criteria and/or BAL criteria.

Hypersensitivity Pneumonitis versus Idiopathic Pulmonary Fibrosis

Forty-nine studies compared the percentage of BAL fluid lymphocytes in patients with HP with that in patients with IPF (15–62): 15 studies enrolled patients with nonfibrotic HP (21, 30–43), 15 studies enrolled patients with fibrotic HP (15–29), and 19 studies did not specify the type of HP (44–62). An initial meta-analysis revealed severe heterogeneity, and the cause of the heterogeneity could not be determined; therefore, outliers were removed (21, 27–29, 39–43, 59–62). For all studies, the mean percentage of BAL fluid lymphocytes was significantly higher for HP than IPF (MD, 30.38%; 95% CI, 26.78–33.99%). In subgroup analyses, the mean difference in the percentage of BAL fluid lymphocytes between nonfibrotic HP and IPF was 34.21% (95% CI, 28.66–39.76%), The mean difference in the percentage of BAL fluid lymphocytes between fibrotic HP and IPF was 20.81% (95% CI, 14.45–27.17%) (Figure 2).

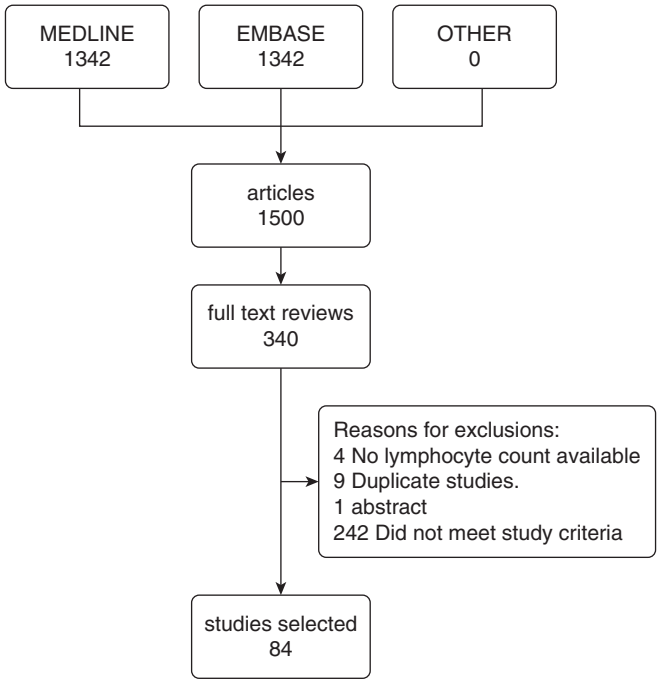

Figure 1. Flow of information diagram.

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A **Before outliers removed**

Study or Subgroup	HP Mean	HP SD	HP Total	IPF Mean	IPF SD	IPF Total	Weight	Mean Difference IV, Random, 95% CI
7.2.1 Non Fibrotic HP vs. IPF								
Bertorelli 1988	44.5	15.1	11	8.8	4.2	7	2.2%	35.70 [26.25, 45.15]
Costabel 1992	78	10	24	15	15	22	2.2%	63.00 [55.56, 70.44]
Drent 1996	55.82	15.98	43	10.78	11.81	82	2.3%	45.04 [39.62, 50.46]
Haslam 1990	45	20	13	7	7	3	2.1%	38.00 [24.55, 51.45]
Hoogsteden 1993	37.2	17.9	9	6.1	8.2	8	2.1%	31.10 [18.10, 44.10]
Kombach 1996	25	5	10	21	4	29	2.3%	4.00 [0.58, 7.42]
Kucejko 2009	39.1	20.8	9	26.7	22.7	7	1.7%	12.40 [–9.22, 34.02]
Magdalena 2003	34.9	7.3	9	17.4	3.2	19	2.3%	17.50 [12.52, 22.48]
Okamoto 2015	82.11	14.45	35	9	8.9	54	2.3%	73.11 [67.77, 78.45]
Papakosta 2014	43.4	22.2	19	19.6	11	38	2.1%	23.80 [13.22, 34.38]
Perez-Arellano 1993	45.47	30.37	5	13.25	17.9	5	1.4%	32.22 [1.32, 63.12]
Pforte 1993	44	5.4	20	31	5.1	20	2.3%	13.00 [9.74, 16.26]
Robinson 1987	50	30	4	17	10.58	7	1.4%	33.00 [2.57, 63.43]
Suga 1999	71.5	21.91	5	8.9	11.91	21	1.8%	62.60 [42.73, 82.47]
Yamashita 2015	52.5	1.83	26	17.5	5.53	14	2.3%	35.00 [32.02, 37.98]
Subtotal (95% CI)			**242**			**336**	**30.7%**	**34.67 [23.01, 46.32]**
Heterogeneity: Tau^2 = 480.51; Chi^2 = 724.66, df = 14 (P < 0.00001); I^2 = 98%								
Test for overall effect: Z = 5.83 (P < 0.00001)								
7.2.2 Fibrotic HP vs. IPF								
Bellanger 2016	46	29	16	12.9	10.4	11	2.0%	33.10 [17.62, 48.58]
Furasawa 2017	29.7	26.5	76	16.8	16.9	107	2.2%	12.90 [6.14, 19.66]
Groot Kormelink 2011	50	19	22	12	7	21	2.2%	38.00 [29.51, 46.49]
Lohtonen 2014	47	30	3	24	19	17	1.2%	23.00 [–12.13, 58.13]
Markart 2009	29.7	21.16	7	6.8	12.8	16	1.9%	22.90 [6.02, 39.78]
Morell 2013	11.1	6.83	19	8.96	4.97	25	2.3%	2.14 [–1.50, 5.78]
Nukui 2019	20.4	20.9	32	3.8	1.8	8	2.2%	16.60 [9.25, 23.95]
Ohshimoto 2009	67	3.6	3	8	6	68	2.3%	59.00 [54.68, 63.32]
Okamoto 2015	27.12	30.49	57	9	8.9	54	2.2%	18.12 [9.86, 26.38]
Pesci 1993	56.5	18.59	15	8.1	5.4	9	2.2%	48.40 [38.35, 58.45]
Przbylskl 2015	40.47	18.74	6	12.82	7.13	18	2.0%	27.65 [12.30, 43.00]
Schmidt 2002	34.44	11.73	5	10.92	10.77	12	2.1%	23.52 [11.57, 35.47]
Sterclova 2009	24.6	26.4	16	30	30.54	8	1.6%	–5.40 [–30.20, 19.40]
Vasakova 2008	23.7	28.3	7	22.3	23.2	7	1.5%	1.40 [–25.71, 28.51]
Williems 2013	28	31.57	11	10.4	10.12	11	1.8%	17.60 [–1.99, 37.19]
Subtotal (95% CI)			**295**			**392**	**29.8%**	**23.25 [10.24, 36.26]**
Heterogeneity: Tau^2 = 596.48; Chi^2 = 450.38, df = 14 (P < 0.00001); I^2 = 97%								
Test for overall effect; Z = 3.50 (P < 0.0005)								
7.2.3 Non Specific HP vs. IPF								
Bargagli 2007	37.2	29.1	7	17.9	11.8	15	1.7%	19.30 [–3.07, 41.67]
Cai 2013	53	23.32	34	9	7.34	54	2.2%	44.00 [35.92, 52.08]
Espoladore 2014	32.63	34.35	15	10.72	6.58	14	1.9%	21.91 [4.19, 39.63]
Fireman 1994	58	12	3	13	13	3	1.8%	45.00 [24.98, 65.02]
Gunther 1999	32.1	23.75	32	14.7	16.2	36	2.2%	17.40 [7.62, 27.18]
Hill 2004	62.2	20.7	115	17.6	7	94	2.3%	44.60 [40.56, 48.64]
Hoogsteden 1989	29.2	20	7	8.4	9.4	18	2.0%	20.80 [5.36, 36.24]
Jara-Palmores 2009	47.3	23.5	64	17.3	15.4	76	2.2%	30.00 [23.28, 36.72]
Kopinski 2011	53.9	6.6	7	13.3	3.7	16	2.3%	40.60 [35.39, 45.81]
Oshima 1999	71.2	9.8	4	7.4	7.58	10	2.1%	63.80 [53.11, 74.49]
Pardo 2001	66	17	10	27	18	9	2.0%	39.00 [23.21, 54.79]
Phelps 2004	67.5	21.8	23	23.6	13.6	26	2.1%	43.90 [33.57, 54.23]
Raulf 1993	19.1	10.9	13	7.9	4.3	18	2.2%	11.20 [4.79, 17.61]
Sherson 1992	58	13.22	7	17	15.49	15	2.1%	41.00 [28.46, 53.54]
Sokhatska 2019	54.5	21.41	53	26.61	11.52	8	2.2%	27.89 [18.04,37.74]
Thomeer 2004	53	29	50	15	16	37	2.2%	38.00 [28.45, 47.55]
Welker 2004	48	18.18	66	12.93	11.64	112	2.3%	35.07 [30.18, 39.96]
Wojtan 2016	60.41	31.81	10	11.38	4.76	6	1.8%	49.03 [28.95, 69.11]
Zissel 2000	34.8	22.2	9	7.1	5.8	11	2.0%	27.70 [12.80, 42.60]
Subtotal (95% CI)			**529**			**572**	**39.5%**	**34.91 [29.10, 40.72]**
Heterogeneity: Tau^2 = 129.85; Chi^2 = 142.46, df = 18 (P < 0.00001); I^2 = 87%								
Test for overall effect; Z = 11.77 (P < 0.00 01)								
Total (95% CI)			**1066**			**1300**	**100.0%**	**31.41 [25.60, 37.21]**
Heterogeneity: Tau^2 = 380.58; Chi^2 = 1390.92, df = 48 (P < 0.00001); I^2 = 97%								
Test for overall effect; Z = 10.60 (P < 0.00001)								
Test for subgroup differences: Chi^2 = 2.64, df = 2 (P = 0.27), I^2 = 24.2%								

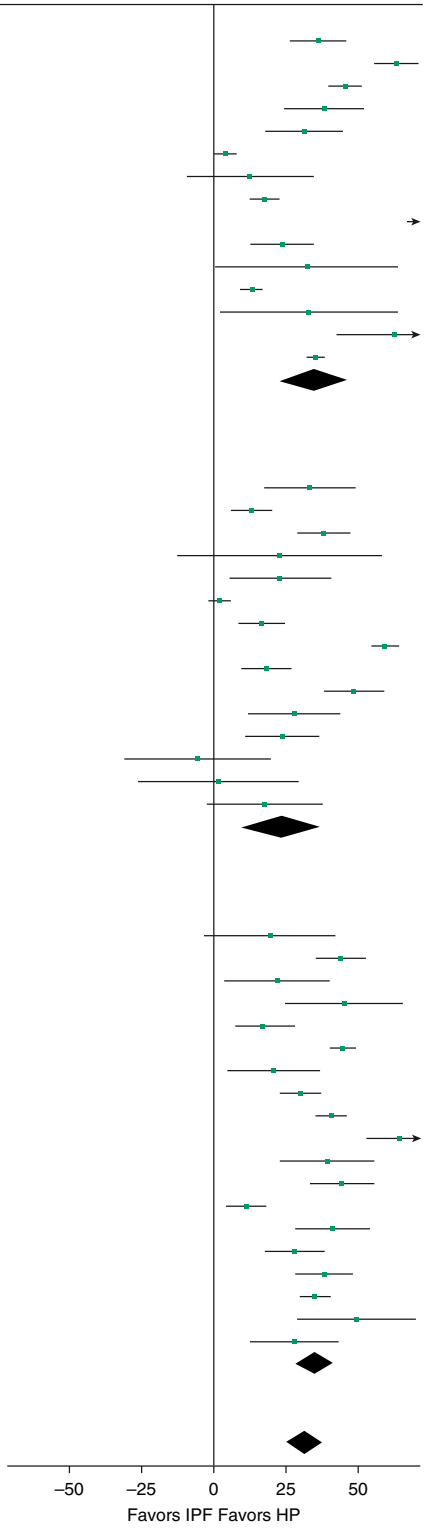

Figure 2. Mean bronchoalveolar lavage lymphocyte percentage in patients with HP compared with patients with idiopathic pulmonary fibrosis. (*A*) Before outliers removed. (*B*) After outliers removed. 95% CI = 95% confidence interval; df = degrees of freedom; HP = hypersensitivity pneumonitis; I^2 = percentage of variation across studies due to heterogeneity; IPF = idiopathic pulmonary fibrosis; IV = inverse variance; SD = standard deviation.

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B

After outliers removed

Study or Subgroup	HP Mean	HP SD	HP Total	IPF Mean	IPF SD	IPF Total	Weight	Mean Difference IV, Random, 95% CI
7.2.1 Non Fibrotic HP vs. IPF								
Bertorelli 1988	44.5	15.1	11	8.8	4.2	7	3.5%	35.70 [26.25, 45.15]
Costabel 1992	78	10	24	15	15	22		Not estimable
Drent 1996	55.82	15.98	43	10.78	11.81	82	4.1%	45.04 [39.62, 50.46]
Haslam 1990	45	20	13	7	7	3	2.8%	38.00 [24.55, 51.45]
Hoogsteden 1993	37.2	17.9	9	6.1	8.2	8	2.9%	31.10 [18.10, 44.10]
Kombach 1996	25	5	10	21	4	29		Not estimable
Kucejko 2009	39.1	20.8	9	26.7	22.7	7	1.7%	12.40 [–9.22, 34.02]
Magdalena 2003	34.9	7.3	9	17.4	3.2	19		Not estimable
Okamoto 2015	82.11	14.45	35	9	8.9	54		Not estimable
Papakosta 2014	43.4	22.2	19	19.6	11	38	3.3%	23.80 [13.22, 34.38]
Perez-Arellano 1993	45.47	30.37	5	13.25	17.9	5	1.0%	32.22 [1.32, 63.12]
Pforte 1993	44	5.4	20	31	5.1	20		Not estimable
Robinson 1987	50	30	4	17	10.58	7	1.1%	33.00 [2.57, 63.43]
Suga 1999	71.5	21.91	5	8.9	11.91	21		Not estimable
Yamashita 2015	52.5	1.83	26	17.5	5.53	14	4.4%	35.00 [32.02, 37.98]
Subtotal (95% CI)			**139**			**171**	**24.8%**	**34.21 [28.66, 39.76]**
Heterogeneity: Tau^2 = 33.53; Chi^2 = 21.55, df = 8 (P = 0.006); I^2 = 63%								
Test for overall effect: Z = 12.08 (P < 0.00001)								
7.2.2 Fibrotic HP vs. IPF								
Bellanger 2016	46	29	16	12.9	10.4	11	2.5%	33.10 [17.62, 48.58]
Furasawa 2017	29.7	26.5	76	16.8	16.9	107	3.9%	12.90 [6.14, 19.66]
Groot Kormelink 2011	50	19	22	12	7	21	3.6%	38.00 [29.51, 46.49]
Lohtonen 2014	47	30	3	24	19	17	0.9%	23.00 [–12.13, 58.13]
Markart 2009	29.7	21.16	7	6.8	12.8	16	2.3%	22.90 [6.02, 39.78]
Morell 2013	11.1	6.83	19	8.96	4.97	25		Not estimable
Nukui 2019	20.4	20.9	32	3.8	1.8	8	3.8%	16.60 [9.25, 23.95]
Ohshimoto 2009	67	3.6	3	8	6	68		Not estimable
Okamoto 2015	27.12	30.49	57	9	8.9	54	3.7%	18.12 [9.86, 26.38]
Pesci 1993	56.5	18.59	15	8.1	5.4	9		Not estimable
Przbylskl 2015	40.47	18.74	6	12.82	7.13	18	2.5%	27.65 [12.30, 43.00]
Schmidt 2002	34.44	11.73	5	10.92	10.77	12	3.0%	23.52 [11.57, 35.47]
Sterclova 2009	24.6	26.4	16	30	30.54	8	1.4%	–5.40 [–30.20, 19.40]
Vasakova 2008	23.7	28.3	7	22.3	23.2	7	1.3%	1.40 [–25.71, 28.51]
Williems 2013	28	31.57	11	10.4	10.12	11	1.9%	17.60 [–1.99, 37.19]
Subtotal (95% CI)			**258**			**290**	**30.9%**	**20.81 [14.45, 27.17]**
Heterogeneity: Tau^2 = 69.87; Chi^2 = 32.49, df = 11 (P = 0.0006); I^2 = 66%								
Test for overall effect; Z = 6.41 (P < 0.00001)								
7.2.3 Non Specific HP vs. IPF								
Bargagli 2007	37.2	29.1	7	17.9	11.8	15	1.7%	19.30 [–3.07, 41.67]
Cai 2013	53	23.32	34	9	7.34	54	3.7%	44.00 [35.92, 52.08]
Espoladore 2014	32.63	34.35	15	10.72	6.58	14	2.2%	21.91 [4.19, 39.63]
Fireman 1994	58	12	3	13	19	3	1.9%	45.00 [24.98, 65.02]
Gunther 1999	32.1	23.75	32	14.7	16.2	36		Not estimable
Hill 2004	62.2	20.7	115	17.6	7	94		Not estimable
Hoogsteden 1989	29.2	20	7	8.4	9.4	18	2.5%	20.80 [5.36, 36.24]
Jara-Palmores 2009	47.3	23.5	64	17.3	15.4	76	3.9%	30.00 [23.28, 36.72]
Kopinski 2011	53.9	6.6	7	13.3	3.7	16	4.2%	40.60 [35.39, 45.81]
Oshima 1999	71.2	9.8	4	7.4	7.58	10		Not estimable
Pardo 2001	66	17	10	27	18	9	2.4%	39.00 [23.21, 54.79]
Phelps 2004	67.5	21.8	23	23.6	13.6	26	3.3%	43.90 [33.57, 54.23]
Raulf 1993	19.1	10.9	13	7.9	4.3	12		Not estimable
Sherson 1992	58	13.22	7	17	15.49	15	2.9%	41.00 [28.46, 53.54]
Sokhatska 2019	54.5	21.41	53	26.61	11.52	8	3.4%	27.89 [18.04, 37.74]
Thomeer 2004	53	29	50	15	16	37	3.5%	38.00 [28.45, 47.55]
Welker 2004	48	18.18	66	12.93	11.64	112	4.2%	35.07 [30.18, 39.96]
Wojtan 2016	60.41	31.81	10	11.38	4.76	6	1.9%	49.03 [28.95, 69.11]
Zissel 2000	34.8	22.2	9	7.1	5.8	11	2.6%	27.70 [12.80, 42.60]
Subtotal (95% CI)			**365**			**420**	**44.3%**	**35.77 [31.96, 39.58]**
Heterogeneity: Tau^2 = 23.24; Chi^2 = 27.55, df = 14 (P = 0.02); I^2 = 49%								
Test for overall effect; Z = 18.40 (P < 0.00001)								
Total (95% CI)			**762**			**881**	**100.0%**	**30.38 [26.78, 33.99]**
Heterogeneity: Tau^2 = 73.87; Chi^2 = 151.11, df = 35 (P < 0.00001); I^2 = 77%								
Test for overall effect: Z = 16.54 (P < 0.00001)								
Test for subgroup differences: Chi^2 = 16.16, df = 2 (P = 0.0003); I^2 = 87.6%								

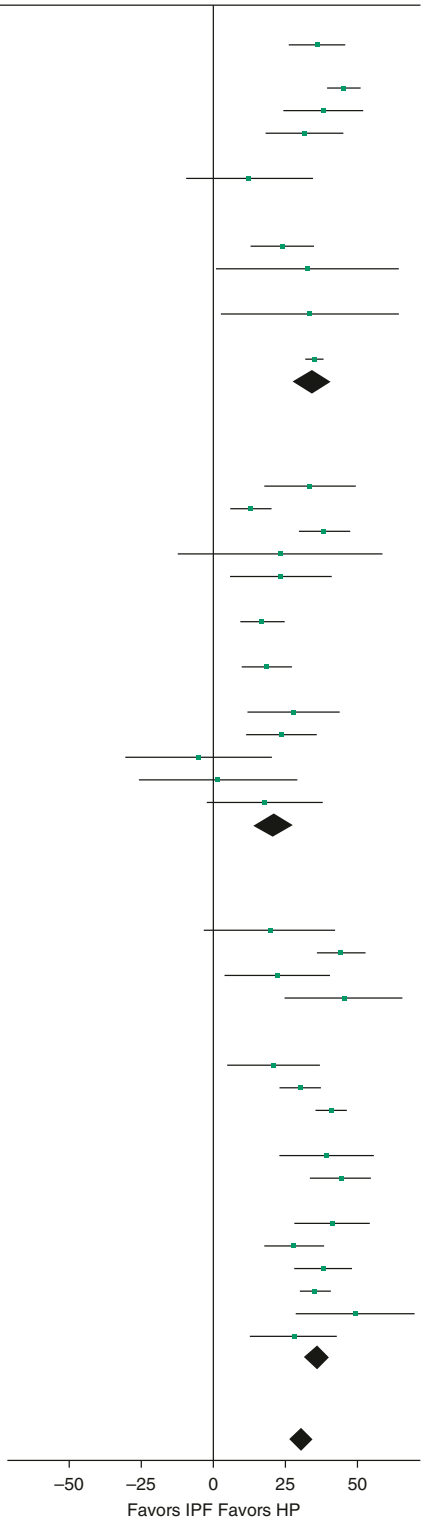

Figure 2. (*Continued*).

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The quality of evidence was judged very low because the body of evidence consisted of case series with risk of bias due to nonconsecutive enrollment of subjects and inconsistent results across studies (Table 1).

Hypersensitivity Pneumonitis versus Sarcoidosis

Seventy-one studies compared the percentage of BAL fluid lymphocytes in patients with HP with that in patients with sarcoidosis (15, 18, 19, 21–23, 25, 29, 30, 32, 33, 36, 40–53, 56–59, 61–96, 97): 25 studies enrolled patients with nonfibrotic HP (21, 30–33, 36, 37, 40–43, 63–76), 17 studies enrolled patients with fibrotic HP (15, 18, 19, 21–23, 25, 29, 74, 77–82, 94, 96), and 29 studies did not specify the type of HP (44–53, 56–59, 61, 62, 83–93, 95, 97). An initial meta-analysis revealed severe heterogeneity, and the cause of the heterogeneity could not be determined; therefore, outliers were removed (21, 33, 40–42, 50, 61, 62, 64, 70, 71, 83, 86, 89, 92, 95). For all studies, the mean percentage of BAL fluid lymphocytes was significantly higher for HP than sarcoidosis (MD, 19.32%; 95% CI, 17.28–21.36%). In subgroup analyses, the mean difference in the percentage of BAL fluid lymphocytes between nonfibrotic HP and sarcoidosis was 24.58% (95% CI, 22.07–27.10%). The mean difference in the percentage of BAL fluid lymphocytes between fibrotic HP and sarcoidosis was 15.68% (95% CI, 11.25–20.10%) (Figure 3).

The quality of evidence was judged very low because the body of evidence consisted of case series with risk of bias due to nonconsecutive enrollment of subjects and inconsistent results across studies (Table 2).

Threshold Percentage BAL Fluid Lymphocytes

Sensitivity and specificity were determined at different thresholds of percentage BAL fluid lymphocytes, and summary receiver operating curves were constructed. To distinguish fibrotic HP from IPF, BAL fluid lymphocyte thresholds of 20%, 30%, and 40% yielded sensitivities of 69%, 55%, and 41%, respectively, and specificities of 61%, 80%, and 93%, respectively, with an AUC of 0.54 (95% CI, 0.51–0.58). To distinguish nonfibrotic HP from IPF, BAL fluid lymphocyte thresholds of 20%, 30%, and 40% yielded sensitivities of 95%, 88%, and 67%, respectively, and specificities of 61%, 80%, and 93%, respectively, with an AUC of 0.75 (95% CI, 0.71–0.78) (Table 3).

To distinguish fibrotic HP from sarcoidosis, BAL fluid lymphocyte thresholds of 20%, 30%, and 40% yielded sensitivities of 69%, 55%, and 41%, respectively, and specificities of 26%, 43%, and 61%, respectively, with an AUC of 0.44 (95% CI, 0.41–0.47).To distinguish nonfibrotic HP from sarcoidosis, BAL fluid lymphocyte thresholds of 20%, 30%, and 40% yielded sensitivities of 95%, 88%, and 76%, respectively, and specificities of 26%, 43%, and 61%, respectively, with an AUC of 0.71 (95% CI, 0.67–0.74) (Table 3).

The low AUCs for fibrotic HP versus IPF and fibrotic HP versus sarcoidosis were a consequence of the high SD in many studies. As a result, the populations of patients with fibrotic HP, patients with IPF, and patients with sarcoidosis also had a high SD, meaning that their distributions were wide and overlapping.

A sensitivity analysis was performed to assess potential inclusion bias (overestimation of sensitivity and specificity due to the inclusion of studies that used BAL fluid lymphocytosis as a diagnostic criterion for HP); the results remained the same, and therefore, the studies were kept in the final analysis.

Table 1. Evidence profile for HP versus IPF

Quality Assessment							Summary of Findings			Quality	Importance
Studies (*n*)	Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other	HP (*n*)	IPF (*n*)	Mean Difference (95% CI)		
All–HP vs. IPF (% lymphocytes)											
36*	NRS	Serious†	Serious‡	None	None	None	762	881	30.38 (26.78–33.99)	⊕OOO Very low	Critical
Fibrotic/chronic HP vs. IPF (% lymphocytes)											
12§	NRS	Serious†	Serious‡	None	None	None	258	290	20.81 (14.45–27.17)	⊕OOO Very low	Critical
Nonfibrotic/acute/subacute HP vs. IPF (% lymphocytes)											
9‖	NRS	Serious†	Serious‡	None	None	None	139	171	34.21 (28.66–39.76)	⊕OOO Very low	Critical
Nonspecified HP vs. IPF (% lymphocytes)											
15¶	NRS	Serious†	Serious‡	None	None	None	365	420	35.77 (26.78–33.99)	⊕OOO Very low	Critical

Definition of abbreviations: 95% CI = 95% confidence interval; HP = hypersensitivity pneumonitis; I^2 = percentage of variation across studies due to heterogeneity; IPF = idiopathic pulmonary fibrosis; NRS = nonrandomized study.
*Studies included References 15–26, 30–38, 44–58.
†Multiple studies had a serious risk of bias due to lack of enrollment of consecutive patients, as shown in the selected studies table.
‡Even after the removal of outliers, the I^2 statistic was >50%.
§Studies included References 15–26.
‖Studies included References 30–38.
¶Studies included References 44–58.

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A — Before outliers removed

Study or Subgroup	HP Mean	HP SD	HP Total	Sarcoid Mean	Sarcoid SD	Sarcoid Total	Weight	Mean Difference IV, Random, 95% CI
8.1.1 Non Fibrotic HP vs Sarcoid								
Agostini 1987	78.2	6.1	8	51.6	10.1	11	1.8%	26.60 [19.29, 33.91]
Agostini 1996	57	5.2	8	47.4	8.8	10	1.9%	9.60 [3.06, 16.14]
Baumer 1998	33.5	23.55	12	16	15.36	41	1.4%	17.50 [3.37, 31.63]
Bertorell 1988	44.5	15.1	11	31.4	13.9	19	1.6%	13.10 [2.21, 23.99]
Cantin 1984	61.4	21.55	11	32.6	20.14	23	1.3%	28.80 [13.64, 43.96]
Cuoto 2015	70.8	13.86	20	34.8	47.31	41	1.3%	36.00 [20.30, 51.70]
Drent 1996	55.82	15.98	43	33.2	19.4	281	1.9%	22.62 [17.33, 27.91]
Fujishima 2004	64.2	18.52	7	28.1	17.63	24	1.3%	36.10 [20.67, 51.53]
Hamagami 1992	55.4	12	16	25.9	10	11	1.8%	29.50 [21.16, 37.84]
Haslam 1990	45	20	13	29	21	23	1.4%	16.00 [2.15, 29.85]
Heron 2009	46.25	14.17	7	47	17	23	1.5%	−0.75 [−13.34, 11.84]
Hoogsteden 1993	37.2	17.9	9	35.7	17.1	10	1.3%	1.50 [−14.28, 17.28]
Kombach 1996	25	5	10	19	4	19	2.0%	6.00 [2.42, 9.58]
Magdalena 2003	34.9	7.3	9	27.3	3.7	18	1.9%	7.70 [2.63, 12.77]
Ohtuska 1994	80.2	8.7	12	41	1.8	12	1.6%	39.20 [27.89, 50.51]
Okamoto 2015	82.11	14.45	35	42	26.53	47	1.7%	40.11 [31.14, 49.08]
Perez-Arellano 1993	45.47	30.37	5	43.11	58.32	5	0.2%	2.36 [−55.27, 59.99]
Pforte 1993	44	5.4	20	37	3.6	20	2.0%	7.00 [4.16, 9.84]
Roblnson 1987	50	30	4	36	13.41	20	0.6%	14.00 [−15.98, 43.98]
Shinohara 2018	43.2	24.3	4	21.8	15.4	11	0.8%	21.40 [−4.09, 46.89]
Suga 1999	71.5	21.91	5	30.1	15.6	16	1.0%	41.40 [20.73, 62.07]
Tanluchi 2009	65.2	4.1	14	40.8	2.2	76	2.0%	24.40 [22.20, 26.60]
Unoura 2011	74.86	42.53	19	42.9	70.17	10	0.3%	31.96 [−15.55, 79.47]
Uzaslan 2003	75	18	14	45	18	11	1.4%	30.00 [15.79, 44.21]
Valenti 1982	58.35	19.43	9	39.15	25.7	20	1.2%	19.20 [2.23, 36.17]
Subtotal (95% CI)			**325**			**802**	**35.2%**	**20.59 [15.61, 25.56]**
Heterogeneity: Tau^2 = 111.55; Chi^2 = 239.86, df = 24 (P < 0.00001); I^2 = 90%								
Test for overall effect: Z = 8.11 (P < 0.00001)								
8.1.2 Fibrotic HP vs. Sarcoid								
Bellanger 2016	46	29	16	27.5	19.1	12	1.1%	18.50 [0.65, 36.35]
Dal 2005	66.8	14.2	12	53.8	14.86	10	1.5%	13.00 [0.78, 25.22]
Delacroix 1985	41	11.4	4	31.38	41.14	17	0.9%	9.62 [−12.90 32.14]
Hamm 1994	67	4.1	10	46	2.7	35	2.0%	21.00 [18.31, 23.69]
Korosec 2007	52	16	17	38	16.75	57	1.7%	14.00 [5.24, 22.76]
Leatherman 1984	65	9.79	6	38	17.46	6	1.3%	27.00 [10.98, 43.02]
Lohtonen 2014	47	30	3	33	19	17	0.5%	14.00 [−21.13, 49.13]
Markart 2009	29.7	21.16	7	21.3	19.05	12	1.1%	8.40 [−10.62, 27.42]
Okamoto 2015	27.12	30.49	57	42	26.53	47	1.6%	−14.88 [−25.84, −3.92]
Pesci 1993	56.5	18.59	15	26.3	16.8	16	1.5%	30.20 [17.70, 42.70]
Przbylski 2015	40.47	18.74	6	36.79	18.84	45	1.3%	3.68 [−12.29 19.65]
Schmidt 2002	34.44	11.73	5	18	10.11	12	1.5%	16.44 [4.67, 28.21]
Sharma 1989	58.66	24.51	3	40.5	18.77	36	0.7%	18.16 [−10.24, 46.56]
Sterclova 2009_1	34.5	36.9	5	17.4	21.1	13	0.5%	17.10 [−17.22, 51.42]
Sterclova 2013	19.5	21.9	14	15.8	8.4	6	1.4%	3.70 [−9.60, 17.00]
Unoura 2011	24.79	25.81	35	42.9	70.17	10	0.3%	−18.11 [−62.43, 26.21]
Vasakova 2008	23.7	28.3	7	19.2	19.4	22	0.9%	4.50 [−17.98, 26.98]
Subtotal (95% CI)			**222**			**373**	**19.9%**	**12.16 [5.82, 18.49]**
Heterogeneity: Tau^2 = 99.25; Chi^2 = 58.64, df = 16 (P < 0.00001); I^2 = 73%								
Test for overall effect: Z = 3.76 (P = 0.0002)								
8.1.3 Non Specific HP vs. Sarcoid								
Bargagli 2007	37.2	29.1	7	30.9	19.7	22	0.9%	6.30 [−16.78, 29.38]
Cal 2013	53	23.32	34	38	19.89	44	1.7%	15.00 [5.20, 24.80]
Costabel 1985	69	15	9	49	23	17	1.3%	20.00 [5.32, 34.68]
Drent 1997	54.3	2.15	54	37.6	1.89	77	2.1%	16.70 [15.99, 17.41]
Espoladore 2014	32.63	34.35	15	26.22	27.67	12	0.9%	6.41 [−16.98, 29.80]
Fireman 1994	58	12	3	48	15	6	1.1%	10.00 [−8.12, 26.12]
Fujimori 2003	71.3	14.4	9	26.3	18	25	1.5%	45.00 [33.24, 56.76]
Godard 1981	68.9	4.8	6	43	5.1	16	2.0%	25.90 [21.32, 30.48]
Gunther 1999	32.1	23.75	32	24.8	16.58	44	1.7%	7.30 [−2.26, 16.88]
Hoogsteden 1989	29.2	20	7	29.4	17.9	20	1.2%	−0.20 [−16.96, 16.56]
Jara-Palmores 2009	47.3	23.5	64	37.9	21.8	88	1.8%	9.40 [2.06, 16.74]
Kopinski 2011	53.9	6.6	7	44.8	2.3	49	2.0%	9.10 [4.17, 14.03]
Milman 1995	38	19.04	6	26	11.66	77	1.3%	12.00 [−3.46, 27.46]
Nagata 1996	67.6	29.2	7	19.2	13.9	17	0.9%	48.40 [25.78, 71.02]
Oshima 1999	71.2	9.8	4	37.8	13.49	17	1.5%	33.40 [21.85, 44.95]
Raulf 1993	19.1	10.9	13	20.5	8	20	1.8%	−1.40 [−8.28, 5.48]
Satake 1993	65.8	19.4	12	37.8	20.5	26	1.4%	28.00 [14.49, 41.51]
Schiidge 2007	54.5	23.7	51	34.1	20.4	237	1.8%	20.40 [13.40, 27.40]
Semenzato 1985	80.2	9.26	7	31.9	21.84	26	1.6%	48.30 [37.46 59.14]
Sherson 1992	58	13.22	7	32	13.07	19	1.6%	26.00 [14.58, 37.42]
Shlota 1987	73.3	8.7	9	48	15.1	16	1.4%	25.30 [11.95, 38.65]
Sokhatska 2019	54.5	21.41	53	40.14	24.57	106	1.8%	14.36 [6.94, 21.78]
Soler 1987	65.9	18.1	20	41.1	16.9	92	1.7%	24.80 [16.15, 33.45]
Sterclova 2015	27.82	25.53	122	22.98	18.13	136	1.9%	4.84 [−0.62, 10.30]
Thomeer 2004	53	29	50	37	24	50	1.6%	16.00 [5.57, 26.43]
Tondell 2015	69.5	2.3	10	31	5.16	83	2.1%	38.50 [36.69, 40.31]
Welker 2004	48	18.18	66	28.4	17.9	239	2.0%	19.60 [14.66, 24.54]
Wojtan 2016	60.41	31.82	10	33.44	21.61	36	1.0%	26.97 [6.02, 47.92]
Zissel 2000	34.8	22.2	9	20.4	15.3	15	1.2%	14.40 [−2.04, 30.84]
Subtotal (95% CI)			**703**			**1622**	**44.9%**	**19.66 [14.72, 24.59]**
Heterogeneity: Tau^2 = 147.47; Chi^2 = 660.51, df = 28 (P < 0.00001); I^2 = 96%								
Test for overall effect: Z = 7.80 (P < 0.00001)								
Total (95% CI)			**1250**			**2797**	**100.0%**	**18.45 [15.60, 21.33]**
Heterogeneity: Tau^2 = 103.01; Chi^2 = 969.21, df = 70 (P < 0.00001); I^2 = 93%								
Test for overall effect: Z = 12.64 (P < 0.00001)								
Test for subgroup differences: Chi^2 = 4.71, df = 2 (P = 0.09), I^2 = 57.5%								

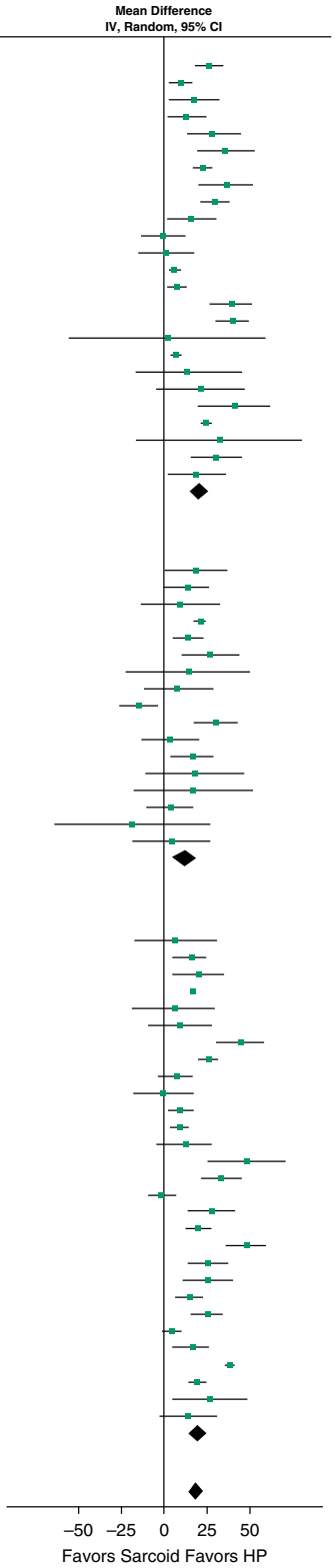

Figure 3. Mean bronchoalveolar lavage lymphocyte percentage in patients with HP compared with patients with sarcoidosis. (*A*) Before outliers removed. (*B*) After outliers removed. 95% CI = 95% confidence interval; df = degrees of freedom; HP = hypersensitivity pneumonitis; I^2 = percentage of variation across studies due to heterogeneity; IV = inverse variance; SD = standard deviation.

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B **After outliers removed**

Study or Subgroup	HP Mean	HP SD	HP Total	Sarcoid Mean	Sarcoid SD	Sarcoid Total	Weight	Mean Difference IV, Random, 95% CI
8.1.1 Non Fibrotic HP vs Sarcoid								
Agostini 1987	78.2	6.1	8	51.6	10.1	11	3.3%	26.60 [19.29, 33.91]
Agostini 1996	57	5.2	8	47.4	8.8	10		Not estimable
Baumer 1998	33.5	23.55	12	16	15.36	41	1.5%	17.50 [3.37, 31.63]
Bertorell 1988	44.5	15.1	11	31.4	13.9	19	2.2%	13.10 [2.21, 23.99]
Cantin 1984	61.4	21.55	11	32.6	20.14	23	1.4%	28.80 [13.64, 43.96]
Cuoto 2015	70.8	13.86	20	34.8	47.31	41	1.3%	36.00 [20.30, 51.70]
Drent 1996	55.82	15.98	43	33.2	19.4	281	4.1%	22.62 [17.33, 27.91]
Fujishima 2004	64.2	18.52	7	28.1	17.63	24	1.3%	36.10 [20.67, 51.53]
Hamagami 1992	55.4	12	16	25.9	10	11	2.9%	29.50 [21.16, 37.84]
Haslam 1990	45	20	13	29	21	23	1.6%	16.00 [2.15, 29.85]
Heron 2009	46.25	14.17	7	47	17	23		Not estimable
Hoogsteden 1993	37.2	17.9	9	35.7	17.1	10		Not estimable
Kombach 1996	25	5	10	19	4	19		Not estimable
Magdalena 2003	34.9	7.3	9	27.2	3.7	18		Not estimable
Ohtuska 1994	80.2	8.7	12	41	18	12		Not estimable
Okamoto 2015	82.11	14.45	35	42	26.53	47		Not estimable
Perez-Arelleno 1993	45.47	30.37	5	43.11	58.32	5	0.1%	2.36 [−55.27, 59.99]
Pforte 1993	44	5.4	20	37	3.6	20		Not estimable
Roblnson 1987	50	30	4	36	13.41	20	0.4%	14.00 [−15.98, 43.98]
Shinohara 2018	43.2	24.3	4	21.8	15.4	11	0.6%	21.40 [−4.09, 46.89]
Suga 1999	71.5	21.91	5	30.1	15.6	16	0.8%	41.40 [20.73, 62.07]
Tanluchi 2009	65.2	4.1	14	40.8	2.2	76	5.3%	24.40 [22.20, 26.60]
Unoura 2011	74.86	42.53	19	42.9	70.17	10	0.2%	31.96 [−15.55, 79.47]
Uzaslan 2003	75	18	14	45	18	11	1.5%	30.00 [15.79, 44.21]
Valenti 1982	58.35	19.43	9	39.15	25.7	20	1.2%	19.20 [2.23, 36.17]
Subtotal (95% CI)			**215**			**643**	**29.6%**	**24.59 [22.07, 27.10]**
Heterogeneity: Tau^2 = 2.92; Chi^2 = 18.04, df = 16 (P = 0.32); I^2 = 11%								
Test for overall effect: Z = 19.19 (P < 0.00001)								
8.1.2 Fibrotic HP vs. Sarcoid								
Bellanger 2016	46	29	16	27.5	19.1	12	1.1%	18.50 [0.65, 36.35]
Dal 2005	66.8	14.2	12	53.8	14.86	10	1.9%	13.00 [0.78, 25.22]
Delacroix 1985	41	11.4	4	31.38	41.14	17	0.7%	9.62 [−12.90 32.14]
Hamm 1994	67	4.1	10	46	2.7	35	5.1%	21.00 [18.31, 23.69]
Korosec 2007	52	16	17	38	16.75	57	2.8%	14.00 [5.24, 22.76]
Leatherman 1984	65	9.79	6	38	17.46	6	1.3%	27.00 [10.98, 43.02]
Lohtoner 2014	47	30	3	33	19	17	0.3%	14.00 [−21.13, 49.13]
Markart 2009	29.7	21.16	7	21.3	19.05	12	1.0%	8.40 [−10.62, 27.42]
Okamoto 2015	27.12	30.49	57	42	26.53	47		Not estimable
Pesci 1993	56.5	18.59	15	26.3	16.8	16	1.8%	30.20 [17.70, 42.70]
Przbylski 2015	40.47	18.74	6	36.79	18.84	45	1.3%	3.68 [−12.29 19.65]
Schmidt 2002	34.44	11.73	5	18	10.11	12	2.0%	16.44 [4.67, 28.21]
Sharma 1989	58.66	24.51	3	40.5	18.77	36	0.5%	18.16 [−10.24, 46.56]
Sterclova 2009_1	34.5	36.9	5	17.4	21.1	13	0.3%	17.10 [−17.22, 51.42]
Sterclova 2013	19.5	21.9	14	15.8	8.4	6	1.7%	3.70 [−9.60, 17.00]
Unoura 2011	24.79	25.81	35	42.9	70.17	10	0.2%	−18.11 [−62.43, 26.21]
Vasakova 2008	23.7	28.3	7	19.2	19.4	22	0.7%	4.50 [−17.98, 26.98]
Subtotal (95% CI)			**165**			**326**	**22.5%**	**15.68 [11.25, 20.10]**
Heterogeneity: Tau^2 = 21.85; Chi^2 = 23.38, df = 15 (P = 0.08); I^2 = 36%								
Test for overall effect: Z = 6.95 (P < 0.00001)								
8.1.3 Non Specific HP vs. Sarcoid								
Bargagli 2007	37.2	29.1	7	30.9	19.7	22	0.7%	6.30 [−16.78, 29.38]
Cal 2013	53	23.32	34	38	19.89	44	2.5%	15.00 [5.20, 24.80]
Costabel 1985	69	15	9	49	23	17	1.4%	20.00 [3.37, 31.63]
Drent 1997	54.3	2.15	54	37.6	1.89	77	5.6%	16.70 [15.99, 17.41]
Espoladore 2014	32.63	34.35	15	26.22	27.67	12	0.7%	6.41 [−16.98, 29.80]
Fireman 1994	58	12	3	48	15	6	1.0%	10.00 [−8.12, 26.12]
Fujimori 2003	71.3	14.4	9	26.3	18	25		Not estimable
Goderd 1981	68.9	4.8	6	43	5.1	16	4.4%	25.90 [21.32, 30.48]
Gunther 1999	32.1	23.75	32	24.8	16.58	44	2.5%	7.30 [−2.28, 16.88]
Hoogsteden 1989	29.2	20	7	29.4	17.9	20	1.2%	−0.20 [−16.96, 16.56]
Jara-Palmores 2009	47.3	23.5	64	37.9	21.8	88	3.3%	9.40 [2.06, 16.74]
Kopinski 2011	53.9	6.6	7	44.8	2.3	49		Not estimabla
Milman 1995	38	19.04	6	26	11.66	77	1.3%	12.00 [−3.46, 27.46]
Nagata 1996	67.6	29.2	7	19.2	13.9	17		Not estimable
Oshima 1999	71.2	9.8	4	37.8	13.49	17		Not estimable
Raulf 1993	19.1	10.9	13	20.5	8	20		Not estimable
Satake 1993	65.8	19.4	12	37.8	20.5	26	1.6%	28.00 [14.49, 41.51]
Schiidge 2007	54.5	23.7	51	34.1	20.4	237	3.4%	20.40 [13.40, 27.40]
Semenzato 1985	80.2	9.26	7	31.9	21.84	26		Not estimable
Sherson 1992	58	13.22	7	32	13.07	19	2.0%	26.00 [14.58, 37.42]
Shlota 1987	73.3	8.7	9	48	15.1	6	1.7%	25.30 [11.95, 38.65]
Sokhatska 2019	54.5	21.41	53	40.14	24.57	106	3.2%	14.36 [6.94, 21.78]
Soler 1987	65.9	18.1	20	41.1	16.9	92	2.8%	24.80 [16.15, 33.45]
Sterclova 2015	27.82	25.53	122	22.98	18.13	136		Not estimable
Thomeer 2004	53	29	50	37	24	50	2.3%	16.00 [5.57, 26.43]
Tondell 2015	69.5	2.3	10	31	5.16	83		Not estimable
Welker 2004	48	18.18	66	28.4	17.9	239	4.2%	19.60 [14.66, 24.54]
Wojtan 2016	60.41	31.82	10	33.44	21.61	36	0.8%	26.97 [6.02, 47.92]
Zissel 2000	34.8	22.2	9	20.4	15.3	15	1.2%	14.40 [−2.04, 30.84]
Subtotal (95% CI)			**524**			**1249**	**47.9%**	**17.68 [14.98, 20.38]**
Heterogeneity: Tau^2 = 13.70; Chi^2 = 43.14, df = 20 (P = 0.002); I^2 = 54%								
Test for overall effect: Z = 12.82 (P < 0.00001)								
Total (95% CI)			**904**			**2218**	**100.0%**	**19.32 [17.28, 21.36]**
Heterogeneity: Tau^2 = 19.17; Chi^2 = 146.52, df = 53 (P < 0.00001); I^2 = 64%								
Test for overall effect: Z = 18.57 (P < 0.00001)								
Test for subgroup differences: Chi^2 = 18.92, df = 2 (P < 0.0001), I^2 = 89.4%								

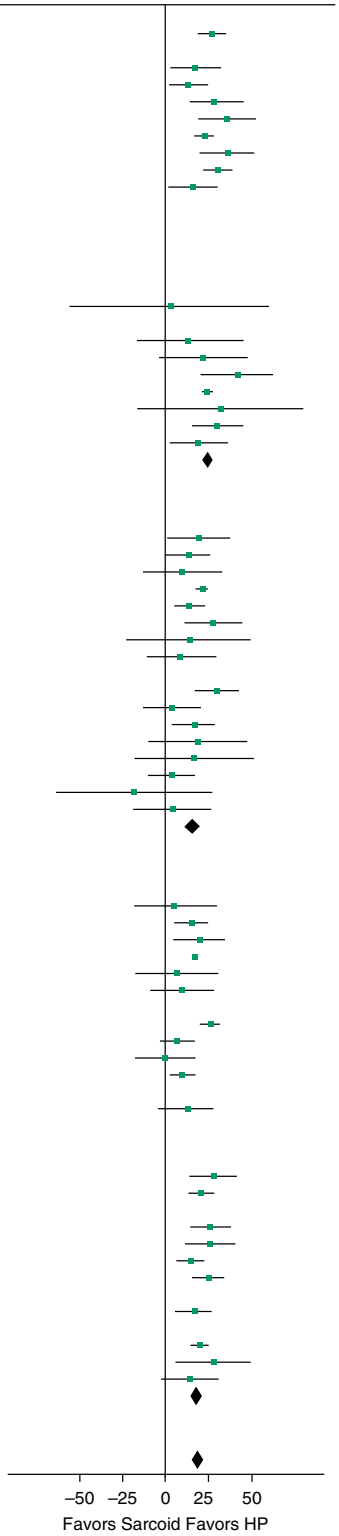

Figure 3. (*Continued*).

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Table 2. Evidence profile for hypersensitivity pneumonitis versus sarcoidosis

Quality Assessment							Summary of Findings			Quality	Importance
Studies (*n*)	Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other	HP (*n*)	Sarcoid (*n*)	Mean Difference (95% CI)		
HP vs. sarcoidosis (% lymphocytes)											
53*	NRS	Serious†	Serious‡	None	None	None	904	2,208	19.32 (17.28–21.36)	⊕OOO Very low	Critical
Fibrotic/chronic HP vs. sarcoidosis (% lymphocytes)											
16§	NRS	Serious†	Serious‡	None	None	None	165	326	15.68 (11.25–20.1)	⊕OOO Very low	Critical
Nonfibrotic/acute/subacute HP vs. sarcoidosis (% lymphocytes)											
17‖	NRS	Serious†	Serious‡	None	None	None	215	643	24.58 (22.07–27.1)	⊕OOO Very low	Critical
Nonspecified HP vs. sarcoidosis (% lymphocytes)											
21¶	NRS	Serious†	Serious‡	None	None	None	524	1,249	17.68 (17.28–21.36)	⊕OOO Very low	Critical

Definition of abbreviations: 95% CI = 95% confidence interval; HP = hypersensitivity pneumonitis; I^2 = percentage of variation across studies due to heterogeneity; NRS = nonrandomized study.
*Studies included References 15, 18, 19, 22, 23, 25, 29–32, 36, 37, 39, 43–49, 51–53, 56–59, 63, 65–69, 72–82, 84, 85, 87, 88, 90, 91, 93, 94, 96.
†Multiple studies had a serious risk of bias due to lack of enrollment of consecutive patients, as shown in the selected studies table.
‡Even after the removal of outliers, the I^2 statistic was >50%.
§Studies included References 15, 18, 19, 22, 23, 25, 29, 74, 77–82, 94, 96.
‖Studies included References 30–32, 36, 37, 43, 63, 65–69, 72–76.
¶Studies included References 39, 44–49, 51–53, 56–59, 84, 85, 87, 88, 90, 91, 93.

Discussion

This is the only systematic review to date that has evaluated the performance of BAL fluid lymphocyte cellular analysis in patients with newly detected ILD and then used the findings to inform clinical practice guideline recommendations of multiple societies. Notably, no relevant randomized trials or controlled observational studies were identified; therefore, case series that reported the percentage of BAL fluid lymphocytes in patients with HP and other ILDs informed the guideline committee.

The systematic review demonstrated that the mean percentage of lymphocytes in the BAL fluid of patients with HP was significantly higher than that in patients with IPF; these differences were greater among patients with nonfibrotic HP than those with fibrotic HP. Similarly, the mean percentage of lymphocytes in the BAL fluid of patients with HP was significantly higher than that in patients with sarcoidosis; again, the difference was greater among patients with nonfibrotic HP than those with fibrotic HP.

These findings are consistent with a study by Drent and colleagues (98), in which sarcoidosis, HP, and IPF could be discriminated from one another using various components of BAL fluid. However, in contrast to our systematic review, Drent used the BAL fluid total cell count and percentages of alveolar macrophages, lymphocytes, polymorphonuclear neutrophils, eosinophils, and plasma cells in a discriminant analysis rather than the

Table 3. Assessment of various diagnostic thresholds of percentage of BAL fluid lymphocytes

	Fibrotic HP vs. IPF		Nonfibrotic HP vs. IPF		Fibrotic HP vs. Sarcoidosis		Nonfibrotic HP vs. Sarcoidosis	
	Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity	Specificity
20% lymphocytes	69%	61%	95%	61%	69%	26%	95%	26%
30% lymphocytes	55%	80%	88%	80%	55%	43%	88%	43%
40% lymphocytes	41%	93%	76%	93%	41%	61%	76%	61%
	AUC 0.54 (95% CI, 0.51–0.58) *N* = 1,238		AUC 0.75 (95% CI, 0.71–0.78) *N* = 1,202		AUC 0.44 (95% CI, 0.41–0.47) *N* = 2,565		AUC 0.71 (95% CI, 0.67–0.74) *N* = 2,529	

Definition of abbreviations: 95% CI = 95% confidence interval; AUC = area under the curve; BAL = bronchoalveolar lavage; HP = hypersensitivity pneumonitis; IPF = idiopathic pulmonary fibrosis.

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percentages of BAL fluid lymphocytes alone. Welker and colleagues (56) also found that the BAL fluid cell differential can provide substantial diagnostic information for conditions like sarcoidosis and those associated with usual interstitial pneumonia, but found that it was less helpful for less frequent conditions like HP.

The systematic review also demonstrated that a diagnostic cutoff of 20% BAL fluid lymphocytes distinguishes nonfibrotic HP from both IPF and sarcoidosis with high sensitivity (few false-negative results), but the specificity was poor (many false-positive results). This suggests that a 20% diagnostic threshold may be helpful in excluding nonfibrotic HP, but it cannot be relied upon to confirm nonfibrotic HP. In contrast, the 20% diagnostic cutoff has poor sensitivity and poor specificity for distinguishing fibrotic HP from either IPF or sarcoidosis. Increasing to a 40% diagnostic threshold increases specificity and decreases sensitivity, but the combination remains suboptimal for distinguishing either nonfibrotic or fibrotic HP from IPF and sarcoidosis. Thus, our findings do not contradict or alter prior guidance, which advocated a threshold of 20% BAL fluid lymphocytes to exclude nonfibrotic HP (98).

These findings are consistent with other studies. Adams and colleagues (100) and Morell and colleagues (101) both reported that diagnostic cutoffs ranging from 20% to 40% had suboptimal sensitivity; neither reported the specificity. Remy-Jardin and colleagues (102) reported that using 18% lymphocytes in the BAL fluid as a diagnostic threshold had sensitivities of 100% and 91% for nonfibrotic and fibrotic HP, respectively; specificity was not reported. Espoladore and colleagues (103) found poor sensitivity and specificity using 25% lymphocytes in the BAL fluid as a diagnostic threshold. More recently, Adderley and colleagues (104) reported that a 20% BAL fluid lymphocyte threshold distinguished chronic HP from other ILDs with a sensitivity and specificity of 68.1% and 64.8%, respectively. Our systematic review similarly found that a 20% BAL fluid lymphocyte threshold distinguished fibrotic HP from IPF with a sensitivity and specificity of 69% and 61%, respectively.

The main strength of this systematic review is that it was done as a part of guideline development. Thus, in addition to all of the usual steps of a systematic review, the methodology team interacted with international experts in the field and patient representatives to ensure that the question was clinically relevant, the outcomes were clinically important, and no studies were missed, whether published or part of the gray literature.

The systematic review has several important limitations. First, there were no randomized trials or observational studies comparing BAL fluid lymphocyte cellular analysis with no BAL fluid lymphocyte cellular analysis; thus, all data and conclusions were from case series with nonconsecutive enrolled patients that measured the percentage of lymphocytes in BAL fluid and compared those values among patients with various ILDs. As a result, there is a serious risk of selection bias. Second, many studies included exposure as a diagnostic criterion. Patients with HP and an unidentified culprit exposure would have been unrepresented in such studies, another potential cause of selection bias. Third, studies categorized patients as having acute, subacute, or chronic HP, whereas the guideline committee categorized patients as having nonfibrotic or fibrotic HP. This was managed by using acute and subacute HP as surrogates for nonfibrotic HP and chronic HP as a surrogate for fibrotic HP, but this approach is inexact. Fourth, studies that did indicate whether a patient had nonfibrotic or fibrotic disease usually made the determination based upon chest radiographs, pulmonary function testing, and duration of symptoms, which is likely insensitive compared with the method most often used today, a CT scan of the chest. Fifth, several studies used BAL fluid lymphocytosis as a criterion for the diagnosis of HP, potentially introducing inclusion bias. This was addressed by removing the studies from the meta-analyses as a part of a sensitivity analysis, which did not reveal definite inclusion bias. Nevertheless, it is possible that inclusion bias existed if the BAL fluid lymphocyte percentage influenced the final diagnosis of HP in studies, even if it was not described. Sixth, most studies did not report details about the site of BAL, the BAL technique, or whether patients were taking steroids or other immunosuppressants, all of which can affect the BAL fluid lymphocyte percentage and could have been a source of heterogeneity. Seventh, most studies did not report on complications associated with the BAL procedure, so the committee was unable to systematically balance the potential benefits of BAL against the potential harms of the procedure. Finally, the analysis to evaluate various thresholds of percentage BAL fluid lymphocytes for distinguishing HP (particularly fibrotic HP) from IPF and sarcoidosis was hampered by the high SD of the individual studies, leading to overlapping populations and an inability to identify an optimal threshold.

The limitations of the systematic review highlight the need for better evidence. Randomized trials or controlled observational studies should be conducted that directly compare BAL fluid lymphocyte cellular analysis with no BAL fluid lymphocyte cellular analysis in patients with ILD. ■

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